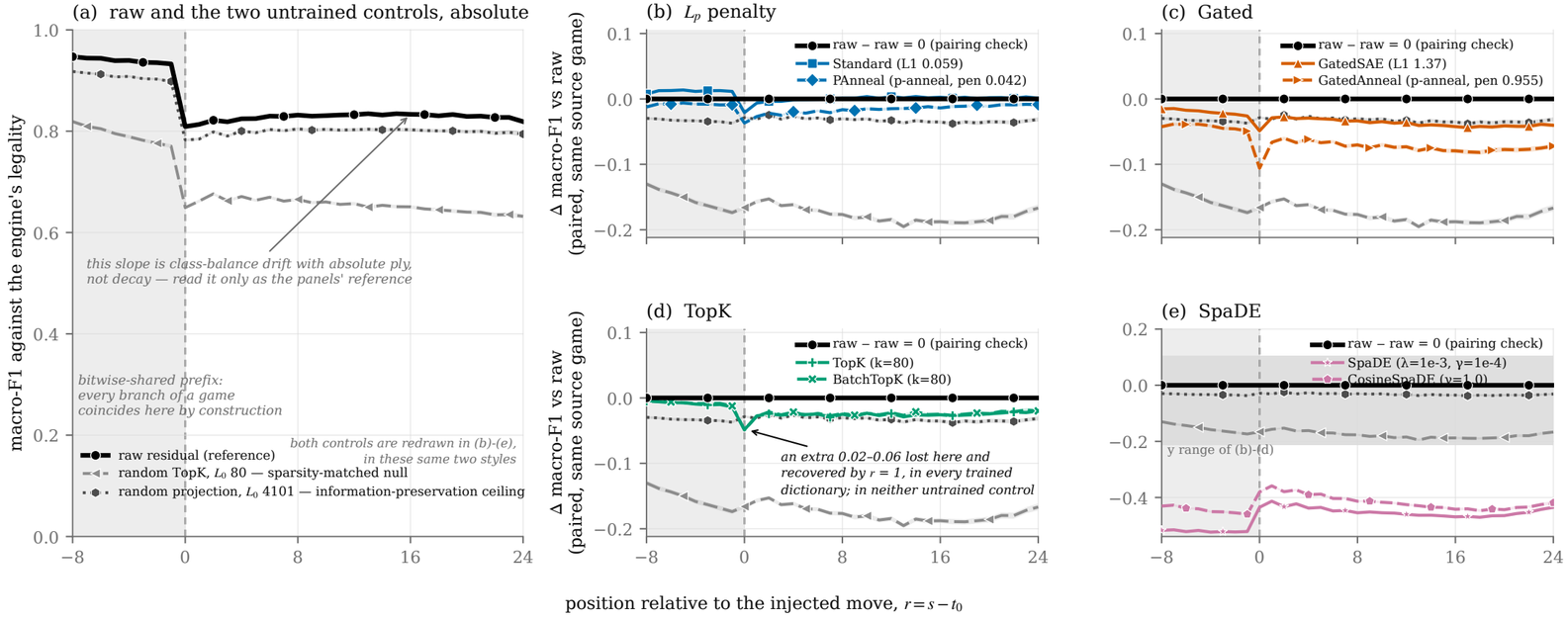
Pooled empty-square violations (the four categories whose square is empty before the move). (a) absolute macro-F1. (b)-(e) every dictionary as a per-game paired difference from raw at the same r, which cancels the class-balance drift exactly because both spaces are scored on the same positions against the same labels. band: 95% bootstrap CI over source games; n = 2000 source games, constant across the window by construction (t_0 round-robin over [10, 34]). CI half-widths are ≤ 0.006 and are drawn but mostly narrower than the lines. Gray span: $r<0$, bitwise identical across branches. (e) has its own y scale.



R1 placement — a move puts one disc of the mover's color on an empty square.
R2 bracketing — the disc must close a gapless line of opponent discs with one of the mover's own.
R3 turn — colors alternate; legality is judged for the mover, not the opponent.